Senior *Blockchain* Architect

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Department: Engineering & *Blockchain* Infrastructure
Reports To: Head of Engineering / CTO
Seniority: Senior / Architecture Leadership

About the Company

We are preparing to establish an independent software development company in Poland, focused on building scalable financial technology and *blockchain*-based applications.

Our planned products include fintech and payment infrastructure, digital asset *trading* platforms, decentralized exchanges (*DEX*), *Web3 wallet*s, *blockchain* integrations, and real estate investment marketplaces.

We aim to build secure, reliable, and scalable software systems that connect traditional financial applications with decentralized technologies.

About the Role

We are looking for an experienced **Senior *Blockchain* Architect** to help design and guide the technical architecture of our *blockchain* and digital asset platforms.

This role combines hands-on *blockchain* engineering with system architecture, technical decision-making, and collaboration across backend, frontend, infrastructure, and product teams.

The ideal candidate has substantial experience designing *blockchain*-based applications, integrating *smart contract*s with traditional backend systems, and building reliable infrastructure for financial transactions.

We are looking for someone who can evaluate different *blockchain* technologies, make practical architectural decisions, and help transform product requirements into scalable technical solutions.

What You'll Do

Blockchain Architecture & System Design

- Design *blockchain* architectures for financial applications, payment platforms, *wallet*s, and digital asset services.
- *Defi*ne how on-chain *smart contract*s interact with off-chain backend systems.
- Evaluate *blockchain* networks, consensus models, execution environments, and transaction processing mechanisms.
- Design scalable systems for *blockchain* transactions, asset transfers, settlement, and reconciliation.
- Establish architecture standards that support maintainability, security, and future integrations.
- Evaluate the trade-offs between *blockchain*-based execution and conventional backend infrastructure.

*Smart Contract*s & Protocol Integration

- Guide the architecture and integration of *smart contract*s across supported *blockchain* networks.
- Collaborate with *smart contract* engineers on design, implementation, testing, and deployment.
- Design contract interaction patterns for financial transactions, *token*ized assets, and decentralized applications.
- Support integrations with *EVM*-compatible networks and other *blockchain* ecosystems where appropriate.
- Review contract designs for security, upgradeability, access control, and operational reliability.
- Evaluate established protocols, SDKs, and open-source infrastructure for potential integration.

Digital Assets, *Wallet*s & Payment Infrastructure

- Architect secure systems for *wallet* connectivity, transaction signing, and digital asset transfers.
- Design reliable transaction lifecycle management, including transaction submission, confirmation, failure recovery, and reconciliation.
- Support *blockchain* payment processing and settlement infrastructure.
- *Defi*ne integration patterns for custodial and non-custodial *wallet* solutions.
- Collaborate on architecture for *DEX* integrations, *token*ized assets, and *Web3* financial services.
- Consider transaction costs, network congestion, finality, and operational risks when designing financial workflows.

Backend & Distributed Systems

- Collaborate with backend engineers to design scalable APIs, microservices, and event-driven architectures.
- *Defi*ne reliable communication between *blockchain* nodes, in*dex*ing systems, backend services, and databases.
- Support transaction monitoring, *blockchain* event processing, and real-time data synchronization.
- Design fault-tolerant workflows using queues, asynchronous processing, retry strategies, and idempotency controls.
- Contribute to infrastructure decisions involving cloud platforms, containerization, and observability.
- Review system architecture for reliability, performance, and operational scalability.

Security & Architecture Governance

- Establish security principles for *blockchain* applications and financial infrastructure.
- Guide secure *smart contract* integration, key management, and transaction authorization.
- Coordinate technical security reviews, threat modeling, and external audits when required.
- Evaluate infrastructure risks involving private keys, signing services, RPC providers, and third-party protocols.
- Ensure that architectural decisions support monitoring, incident response, and recovery procedures.
- Work with relevant stakeholders to incorporate applicable compliance and operational requirements into system designs.

Technical Leadership & Collaboration

- Provide architectural guidance to *blockchain*, backend, frontend, and DevOps engineers.
- Review technical designs, system integrations, and critical implementation decisions.
- Mentor engineers and encourage consistent engineering practices.
- Work closely with product and engineering leadership to evaluate technical feasibility.
- Translate complex *blockchain* concepts into understandable architectural recommendations.
- Participate in technical planning, architecture reviews, and implementation discussions.
- Remain hands-on through prototyping, code reviews, and implementation of technically challenging components.

Required Skills & Experience

- 7+ years of professional software engineering experience, including substantial *blockchain* engineering or architecture experience.
- S*trong* understanding of *blockchain* fundamentals, distributed *ledger*s, transaction processing, consensus, and execution models.
- Proven experience designing and delivering production *blockchain* applications or infrastructure.
- Solid understanding of *EVM*-compatible *blockchain*s, *smart contract*s, and *blockchain* integration patterns.
- Experience with *Solidity* and at least one general-purpose backend language such as Go, *Rust*, Java, Python, or TypeScript.
- S*trong* understanding of distributed systems, APIs, microservices, asynchronous processing, and database architecture.
- Experience integrating *blockchain* networks with backend services and external applications.
- Knowledge of *wallet* infrastructure, digital asset transactions, transaction signing, and *crypto*graphic principles.
- Familiarity with *blockchain* development frameworks, node infrastructure, RPC interfaces, and in*dex*ing solutions.
- Understanding of *blockchain* security, *smart contract* vulnerabilities, and secure system design.
- Experience making architectural decisions, reviewing technical designs, and supporting engineering teams.
- S*trong* problem-solving, communication, and technical documentation skills.
- Ability to work effectively in an international, distributed engineering environment.

Nice to Have (Differentiators)

- Experience designing decentralized exchange or *DeFi* infrastructure.
- Knowledge of account abstraction, multisignature *wallet*s, MPC, or custody architecture.
- Experience with cross-chain messaging, bridges, or interoperability protocols.
- Familiarity with *Solana*, *Rust*-based *blockchain* ecosystems, or alternative execution environments.
- Experience with Layer 2 networks, rollups, zero-knowledge systems, or *blockchain* scaling solutions.
- Background in financial services, payment processing, digital asset custody, or transaction settlement.
- Experience with event-driven *trading* systems, *blockchain* in*dex*ers, or real-time financial data pipelines.
- Familiarity with AWS, Kubernetes, Docker, Terraform, and CI/CD pipelines.
- Contributions to open-source *blockchain* protocols, developer tools, or infrastructure libraries.

- Previous experience mentoring engineers or leading architecture across multiple engineering teams.

Education

- Bachelor's or Master's degree in Computer Science, Software Engineering, Computer Engineering, Mathematics, or a related technical discipline.
- Equivalent professional experience and demonstrated technical achievements are equally valued.

What Success Looks Like

First 3 Months

- Understand the planned product architecture and technical requirements.
- Review *blockchain* network options and major system integration decisions.
- Establish initial architecture principles for *blockchain*, backend, and transaction infrastructure.
- Identify technical risks, security considerations, and implementation priorities.
- Collaborate with engineers on initial prototypes and foundational components.

Within 6 Months

- Guide the delivery of core *blockchain* integration components.
- Establish reliable transaction processing, monitoring, and recovery patterns.
- Improve architectural consistency across engineering teams.
- Support secure *wallet* and digital asset integration workflows.
- Contribute to the development and validation of the platform's technical roadmap.

Long-Term

- Help establish a scalable and secure *blockchain* infrastructure foundation.
- Support the evolution of the company's fintech, digital asset, and decentralized application platforms.
- Reduce architectural complexity and improve engineering reliability.
- Serve as a *trust*ed technical decision-maker for complex *blockchain* systems.

Why Join Us?

- Help shape the architecture of a new independent software company from an early stage.
- Work across fintech, *blockchain*, digital assets, and modern financial infrastructure.

- Contribute to meaningful technical decisions with opportunities for hands-on engineering.
- Collaborate with an international, remote engineering team.
- Explore *blockchain* technologies based on practical product requirements rather than unnecessary complexity.
- Choose from flexible full-time, part-time, or contract collaboration arrangements.

Ideal Candidate

We are looking for an experienced *blockchain* engineer who has progressed into system architecture while maintaining s*tron*g hands-on technical skills.

You should be comfortable designing complex *blockchain* integrations, evaluating trade-offs between decentralized and centralized components, and guiding engineers through difficult technical decisions.

Experience delivering production systems is more important than familiarity with every *blockchain* protocol or programming language.

The s*tron*gest candidate will combine *blockchain* architecture expertise, distributed systems engineering, security awareness, and practical technical leadership.